

"Do not write anything on question-paper except Roll Number, otherwise it shall be deemed as an act of indulging in unfair means and action shall be taken as per rules."

BE II (ME)

Roll No.

3

MD - I

SECOND B.E. EXAMINATION - 2024

MECHANICAL ENGINEERING

(IV SEMESTER CBCS)

4 ME 44 A (ME) : MACHINE DESIGN I

Time - Three Hours

Maximum Marks - 70

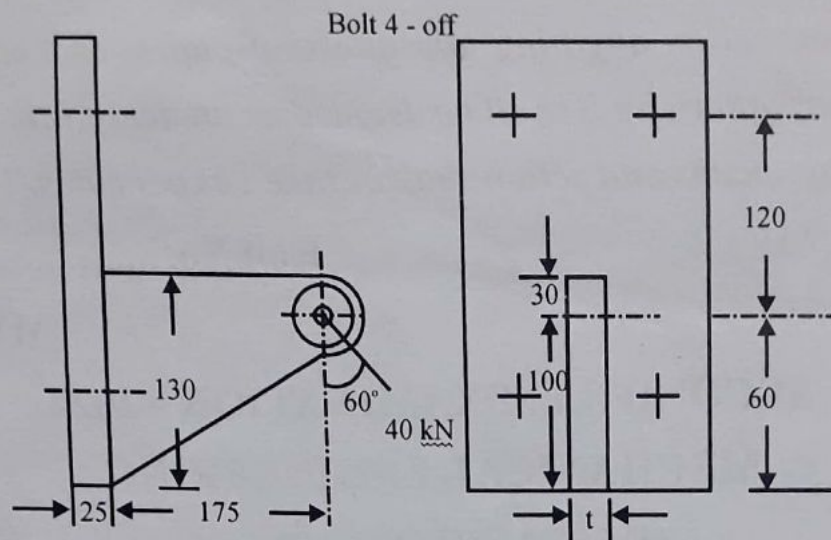
- Note:-** (i) Attempt any **FIVE** questions.
(ii) Answer should be to the point.
(iii) All questions carry equal marks.
(iv) Use of design data hand book is permitted.

1. (a) Write a short note on bolts of uniform strength. 7
(b) Define stress concentration and explain the ways with diagrams the way stress concentration is mitigated. 7
2. Determine the size of the bolts and the thickness of the arm for the bracket as shown in Fig. which carries a load of 4 KN. The material of the bracket is cast steel and bolts are made of mild steel. 7

J-1765

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(Contd.)



All dimensions are in mm

14

3. Design a Knuckle joint to transmit 150 kN. The design stresses may be taken as 75 MPa in tension, 60 MPa in shear and 150 MPa in compression. 14

4. A V-belt drive is to transmit 14.7 kW to a compressor. The motor speed is 1150 rpm. And the compressor pulley runs at 400 rpm. Determine size and number of V-belts required. 14

5. Design a Bell-crank lever of I-section to carry a load of 1000 N. Take length of load arm and effort arm are 250 mm and 400 mm respectively. Also design the fulcrum pin assuming suitable connection type. 14

6. A locomotive semi-elliptical laminated spring has an overall length of 1 m and sustains a load of 70 kN at its centre. The spring has 3 full length leaves and 15 graduated leaves with a central band of 100 mm width. All the leaves are to be stressed to 400 MPa. When fully loaded. The ratio of the total spring depth to that of width is 2.

$E = 210 \text{ kN/mm}^2$. Determine:-

1. The thickness and width of the leaves
2. The initial gap that should be provided between the full length and graduated leaves before the band load is applied.
3. The load exerted on the band after the spring is assembled

14

7. A cast iron flange coupling for a 5 cm diameter shaft is fastened to it by means of rectangular sunk key and is provided with 4 bolt holes at circle diameter of 15 cm for fastening it to other half of the coupling. Making suitable assumptions and assuming that the strength of the key & bolts is 75% of that of the material of the shaft calculate the dimensions of the key, bolts and coupling.

14

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BE II (ME)

Roll No. 23UMEC1159

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FM

SECOND B.E. EXAMINATION - 2024
MECHANICAL ENGINEERING
(IV SEMESTER)

4ME41A (ME) : FLUID MECHANICS

Time - Three Hours

Maximum Marks - 70

- Note:- (i) Attempt any **FIVE** questions.
(ii) Answer should be to the point.
(iii) All questions carry equal marks.

1. (a) Write any three limitations of piezometer. 3
- (b) Derive an expression for hydrostatic force acting on a submerged plane surface. 6
- (c) What do you understand by meta center of a floating body? Define metacentric height. 5
2. (a) Derive continuity equation in three dimensions for Cartesian Coordinates for general flow. Simplify it for various cases of flow. 8

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(Contd.)

~~(b)~~ Define and distinguish among stream line, path line and streak line. 6

3. (a) Derive Euler's equation of motion along a streamline for an ideal fluid stating clearly all the assumptions. 7

~~(b)~~ What is the principle of Venturimeter? Explain. Derive an expression for discharge through a Venturimeter. Why actual discharge is less than theoretical discharge? 7

~~4. (a)~~ Using Buckingham π theorem, derive expression (functional relationship) for force F on the propeller of an aircraft. It depends on forward speed of aircraft V , density of air ρ , Viscosity of air J^e , diameter of propeller D , and speed of rotation of propeller N .

i.e. $F = f(V, \rho, J^e, D, N)$ 10

~~(b)~~ Define and explain Froude number and Weber number, Mention the application of fluid flow where these numbers are applicable. 4

~~5. (a)~~ Derive Darcy's Weisbach formula for calculating loss of head due to friction in pipe. 7

~~(b)~~ Derive formula for calculating head loss due to sudden enlargement. 7

~~6. (a)~~ Derive the expression for velocity distribution for viscous flow through circular pipe. Sketch the velocity distribution and shear stress distribution across a section of the pipe. 7

(b) What do you mean by Reynolds stress? Explain Prandtl mixing length hypothesis and derive relationship between mixing length and shear stresses. 7

X. (a) For the velocity profile in laminar boundary layer as

$$\frac{u}{U} = \frac{3}{2} \left(\frac{y}{\delta} \right) - \frac{1}{2} \left(\frac{y}{\delta} \right)^3$$

Find momentum thickness in terms of δ . Using Von-Korman integral momentum equation prove that

$$\delta/x = \frac{4.64}{\sqrt{Re_x}}$$

where symbols have their usual meanings. 8

(b) What are the causes of separation in boundary layer and how can separation be controlled? 6

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Roll No.1159.....

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MCM

SECOND B.E. EXAMINATION - 2024
MECHANICAL ENGINEERING
(IV SEMESTER)
4ME42A (ME) : METAL CUTTING &
METROLOGY

Time - Three Hours

Maximum Marks - 70

- Note:- (i) Attempt any **FIVE** questions.
(ii) Answer should be to the point.
(iii) All questions carry equal marks.

1. (a) Explain geometry and designation of single point cutting tool. 7
(b) What do you mean by orthogonal cutting? Derive expression for shear angle in metal cutting. 7
2. During turning a carbon steel rod of diameter 160mm by a tool of geometry 0,10,8,9,15,75,0 at speed of 315 rpm, feed of 0.16 mm/rev. and 2.5mm depth of cut, the following observations were made:

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i

(Contd.)

Tangential component of cutting force, $F_c = 500 \text{ N}$

Axial component of cutting force, $F_x = 200 \text{ N}$

Chip thickness, $t_c = 0.48 \text{ mm}$

For the above machining operation determine the values of

- (i) Friction force F , Normal force N at the tool - chip interface.
- (ii) Shear force F_s and normal force F_N at the chip-workpiece interface.
- (iii) Cutting Power in kW
- (iv) Yield strength of the work material under this machining condition.
- (v) Draw merchant circle diagram for above machining condition. 14

3. (a) How does a cutting tool fail? Discuss the flank and crater tool wear mechanism. 7

(b) In a certain machining operation with a cutting speed of 50 m/min, tool life of 45 minutes was observed. When the cutting speed was increased to 100 m/min, the tool life decreased to 10 minutes. Estimate the cutting speed for maximum productivity, if tool change time is 2 minutes. 7

4. Define the terms tolerances, limits and fits with reference to dimensional measurement. Calculate the dimensions of plug and ring gauges to control the production of 50mm shaft and hole pair of H_7/d_8 as per IS specifications. The following assumptions may be made:

50mm lies in diameter step of 30-50mm. Upper deviation for 'd' shaft is $-16D^{0.44}$ and lower deviation for hole 'H' is zero. The tolerance unit 'i' in microns is $=0.45(D)^{1/3} + 0.001$ D and $IT7 = 16i$ and $IT8 = 25i$. Also indicate the limits and tolerances on a diagram. 14

5. (a) Explain Optical Interferometry. 7

(b) Explain the procedure for measurement of screw threads, including the major diameter, minor diameter, effective diameter, pitch, best wire size and thread angle. 7

6. (a) What do you mean by acceptance sampling? Explain single and multiple sampling plans with example. 7

(b) A mobile charger supplier drawn randomly constant sample size of 500 charger every day for quality control test. Defects in each charger are recorded during testing. Based on the given data, draw the appropriate control chart and comment on the state of control. 7

Lot	1	2	3	4	5	6	7	8	9	10
Sample size	500	500	500	500	500	500	500	500	500	500
No. of defects in sample	12	14	16	18	16	14	12	12	32	16

7. Write short note on any two of the following:

(a) Pneumatic comparator

(b) Assignable and unassignable causes of variability in quality

(c) Measurement of gears.

2x7 = 14

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BE II (ME)

Roll No.11.59.....

Kin. & Dyn. of Mac. (M)

SECOND B.E. EXAMINATION - 2024

MECHANICAL ENGINEERING

(IV SEMESTER CBCS)

**4 ME - 43 A : KINEMATICS & DYNAMICS OF
MACHINES (M)**

Time - Three Hours

Maximum Marks - 80

Note:- (i) Attempt any FIVE questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

(iv) Assume suitable data, if not given and mention it clearly.

(v) Draw neat sketches, wherever necessary.

(vi) Q.No. 1 complete solution (including calculations) is to be given in the drawing sheet.

- 1. A cam with 30 mm minimum radius is rotating clockwise at 1200 rpm to give the follower motion to a roller follower of 20 mm diameter. (i) Lift is 25 mm; (ii)**

J-1764

1

(Contd.)

follower rise during 120° cam rotation with simple harmonic motion; (iii) follower dwell for 60° cam rotation; (iv) follower to return during 90° cam rotation with uniform acceleration and deceleration; (v) follower to dwell for remaining period. Draw and describe the profile of the cam and determine the maximum velocity and maximum acceleration during rising and return stroke, also draw displacement, velocity and acceleration diagram for the same.

16

2. (a) A Porter governor has equal links of 250 mm long pivoted at the axis. The weight of each ball is 3 N and the weight of sleeve is 14 N. The radius of the ball is 150 mm when governor begins to lift and 200 mm at the maximum speed. If the friction at the sleeve is equivalent to 1.5 N, determine the maximum and minimum speeds. 8
- (b) What is meant by the term stability of a governor? Derive the necessary condition of stability for the centrifugal governor. What are the limitations of a watt governor? 8
3. (a) A differential band brake is used to retard a shaft carrying a flywheel weighing 3 kN with radius of gyration 350 mm. If the brake drum diameter is 250 mm and coefficient of friction is 0.15, determine the following (i) the braking torque when an effort of 150 N is applied at the end of the lever. (ii) Number of revolutions of the flywheel before it comes to rest. (iii) Time taken by the flywheel to come to rest.

J-1764

2

(Contd.)

The drum rotates at 500 rpm in clockwise direction and the layout of the brake is as given in Figure 1.

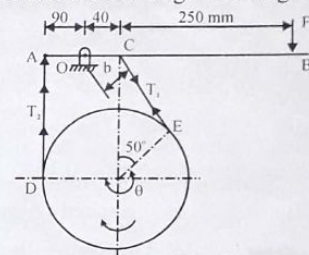


Figure 1.

8

- (b) Derive the formula for ratio of tensions in a band and block brake if the brake consists of n number of blocks. 8
4. (a) Sketch and describe one form of transmission dynamometer. State clearly what dimensions and measurements would have to be taken and explain how the power transmitted may be calculated? 8
- (b) A car moving on a level road at a speed 50 km/hour has a wheel base 2.8 m, distance of C.G. from ground level 600 mm, and the distance of C.G. from rear wheels 1.2 m. Find the distance travelled by the car before coming to rest when brakes are applied, (i) to the rear wheel, (ii) to the front wheel, (iii) to all the four wheels. The coefficient of friction between the tyres and the road may be taken as 0.4. 8

J-1764

3

(Contd.)

5. (a) A motorcycle and its rider together weighs 2kN and their combined centre of gravity is 550 mm above the road when motorcycle is upright. Each wheel is of 580 mm diameter and has a moment of inertia of 1.0 kg-m^2 . The moment of inertia of rotating parts of engine is 0.15 kg-m^2 . The engine rotates at 5 times the speed of the vehicle and in the same sense. Determine the angle of heel necessary when motorcycle is taking a turn over a track of 35m radius at a speed of 60 mph. 8

(b) Explain the effect of gyroscopic couple on a Naval Ship during (i) Steering (ii) Pitching and (iii) Rolling and draw neat diagrams for each case. 8

6. (a) An external force of 10 N is acting horizontally on the rocker link, 30 mm from the point D as shown in Figure 2. Find the amount of torque to be applied to the crank AB to keep the mechanism in static equilibrium and draw the free body diagram. Here, AD = 80mm, AB = 30 mm, BC = 70 mm, CD=50 mm. 10

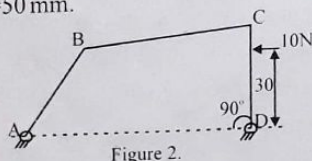


Figure 2.

J-1764

4

(Contd.)

- (b) Explain the concept of free body diagrams using a suitable example. 6

7. Write short notes (any four). 4x4 = 16

(a) Describe various types of cams and follower with neat sketches.

(b) What is the function of a governor? How is it different from that of a flywheel?

(c) Explain self-energising brakes.

(d) What is the principle of virtual work? Explain.

(e) Explain the gyroscopic effect on two-wheeled vehicles.

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BE II (ME)

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IE

SECOND B.E. EXAMINATION - 2024

MECHANICAL ENGINEERING

(IV SEMESTER)

4ME45 A (ME) : INDUSTRIAL ENGINEERING

Time - Three Hours

Maximum Marks - 80

- Note:- (i) Attempt any **Five** questions.
(ii) Answer should be to the point.
(iii) All questions carry equal marks.

1. ~~(a)~~ What is Demand Curve? Explain it briefly how it is different from supply curve? 8
~~(b)~~ What do you understand by Demand Forecasting? Classify it and explain qualitative techniques briefly? 8
2. (a) The sales of Car in a show room in four consecutive months is 70, 68, 82, 95 respectively with a smoothing constant of $\alpha = 0.4$ find the forecast for next month. 8

J-1766

1

(Contd.)

- (b) What should a industry do to maintain healthy industrial relation between management and unions? Explain it briefly? 8
3. (a) What elements of costing defines? Classify the costs with appropriate examples? 8
- (b) What is Depreciation? Write all the methods which can be used to calculate the depreciation. 8
4. (a) What is Budget? Draw the framework for the budget? How many types of budget we can used for industry? 8
- (b) Define value analysis? Explain the procedure that can be used in value analysis? 8
5. (a) Define Work Sampling? Write the applications of it, advantages and disadvantages? 8
- (b) What do you understand by maintenance? Briefly classify it with examples. 8
6. (a) Various charts that can be use for recording techniques? With example. 8
- (b) Explain the principles of motion economy? Briefly. 8
7. Write short notes (any four)
- (a) Wage payment
- (b) Collective bargaining
- (c) System reliability
- (d) Waste management
- (e) Product life cycle
- (f) Productivity
- 4X4=16

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BE II (PI)

7

Open Elective - I

SECOND B.E. EXAMINATION - 2024

**PRODUCTION AND INDUSTRIAL
ENGINEERING**

(IV SEMESTER)

4OE61A(PI) : MANUFACTURING SCIENCE

Time - Three Hours

Maximum Marks - 100

- Note:- (i) Attempt any **Five** questions.
(ii) Answer should be to the point.
(iii) All questions carry equal marks.

1. (a) "Two plates were welded together and then the strength of the joint was tested. It was found that the weld was stronger than either of the plates." Do you think that above statement is incorrect? Comment, giving valid reasons. 10
- (b) Describe the types of flames obtained in an oxy-acetylene gas welding process giving the applications. 10

J-1815

1

(Contd.)

2. (a) What are the various elements that comprise the Gating system? Describe the objectives of gating system in any casting. 10
- (b) Explain the various types of Pattern allowances. What is the necessity of importing allowances to patterns for a Sand-casting? 10
3. (a) Briefly describe the various steps required to manufacture a powdered metal product. 10
- (b) Enumerate the advantages, applications, and limitations of powder metallurgy as a manufacturing process. 10
4. (a) Explain the operations that are normally employed in forging. Also explain how forging improves the mechanical properties of components. 10
- (b) Enlist various processes of metal working and describe in detail any two processes. 10
5. (a) Explain the term 'Machine tool' and how is it different from machine? 10
- (b) How is grinding different from other machining operations? Explain its applications in view of its capabilities. 10
6. (a) Describe the reasons why the unconventional machining processes are used in industry. 10

- (b) Distinguish between bending and drawing in sheet - metal operations. Why is blank holding necessary in a sheet - metal drawing operation? 10

7. Write short notes on any two of the following :-

- (i) Sintering
(ii) Centre lathe
(iii) Sand Moulding

10X2 = 20